Hybrid Quantum-Classical Method for Bank Account Fraud Detection using Quantum Encoding and Quantum Machine Learning

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2024 IEEE MIT Undergraduate Research Technology Conference (URTC) | 979-8-3315-3100-3/24/$31.00 ©2024 IEEE | DOI: 10.1109/URTC65039.2024.10937533

***Abstract*—Fraud detection has attracted interest because it can ensure corporate sustainability, minimize losses, and catch fraudsters. We introduce a novel classification/terminology of Devastating Evolutionary Dynamic Entities (DEDE), where fraud and its sub-types fall under. The paper emphasizes the ineffectiveness of traditional fraud detection methods and calls for the adoption of state-of-the-art fraud detection models immediately. Due to hardware constraints, a hybrid quantum-classical setting was used. Although hybrid methods do not outperform classical methods under current hardware limitations, they still perform moderately well under limited samples, hardware, and features. Overall, we believe that quantum computing has great potential for fraud detection.**

***Index Terms*—Hybrid Quantum-Classical, Machine Learning, Quantum Machine Learning, Financial Fraud Detection, Devastating Evolutionary Dynamic Entity**

I. Introduction

Fraud detection is a critical area of research in financial and security industries. Traditional rule-based decision algorithms, while widely used, often fall short in adaptability and efficiency. These traditional methods are often costly, requiring lots of manpower and training, and become increasingly harder to detect fraud as fraud Patterns change over time. Advanced artificial intelligence models such as XGBoost have demonstrated significant improvements in detection, but they are not without limitations. Looking forward, quantum computing presents a promising but currently impractical solution due to hardware constraints. This paper talks about the shortcomings of conventional rule-based algorithms to detect fraud and compares the accuracy difference between using a classical state-of-the-art method and using a Hybrid Quantum-Classical System where preprocessing is done on classical states and models on the quantum state. to detect fraud due to current hardware limitations.

As financial transactions increasingly take place online, financial fraud continues to grow and have many implications for society. Consequently, fraud detection has become a major concern for financial institutions and other organizations alike, leading many organizations to develop advanced anti-fraud programs.

Our understanding of fraud depends on recognizing its evolutionary capabilities. This requires adopting certain practices and reasoning from those who know how to classify evolutionary growth: early-stage seed investors. Just as investors recognize how startups evolve in response to market forces, we can better predict fraud's risks by understanding its evolutionary patterns. Using metrics such as technological advancements, projected yearly growth, and revenue models, these investors support startups based on their long-term potential and evolutionary growth. While we draw from their methods and analytical approaches, it is important to note that we are not adopting their mindset or broader philosophies, but rather focusing on how they define and measure growth through financial terms.

Under the lens of evolutionary growth, we introduce a novel classification system of Devastating Evolutionary Dynamic Entities (DEDE's). In doing so, we plan to create a metric that measure financial evolutionary harm, and hope that future researchers will continue to reference this term and improve upon the metric.

As of right now, we define a DEDE as an entity that can

1) exhibit continuous evolutionary growth and continues to grow with advancing technology. This can be measured by various metrics such as the annual global/regional increase in losses, cases, or people affected
2) render many currently systems/technology near ineffective
3) Cause financial losses of at least 250 billion globally or proportionally as much of an impact regionally.

We have classified fraud as a whole and even sub-types (payment, identity theft, and corporation fraud) as part of the DEDE's archetype. We outline each of conditions/threshold that fraud has surpassed. As a metric, we hope that future researchers take this terminology and build upon it creating a national scaling system of evolutionary capability.

The first condition is that the entity of fraud must have evolutionary growth that grows with advancing technology. Fraud inherently is challenging to detect and is advancing as methods of fraud become more effective [1]. Technology and time further influence the power of fraud as new varieties

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and methods are uncovered. For example, fraudsters have recently been vitalizing various new methods of fraud such as ”defrauding”, which aims to make current fraud transactions look normal so as not to be flagged [2].

Fraud’s evolutionary growth can be further highlighted for credit card fraud, a single type of fraud, which increased by losses of 6.81 billion USD, rising from 21.84 billion loss in 2015 to 28.65 billion loss in 2019. This means an average increase of 1.73 billion USD loss increase per year. [3].

LOSS CHART

Months	6 or less	7-12	13-36	36 or more
Loss	$30,000	$120,000	$225,000	$650,000+

Fig. 1. Organizational Losses in USD for various time increments

The loss continues to grow for every month the fraud goes unnoticed. Figure 1 shows the USD loss for fraud detection models. It shows that between 6 months or less, a person is capable of losing 30,000 USD. Between 7 and 12 months, you can lose 120,000, and so on. [1].

The second condition is that current technology should be ineffective at dealing with conventional fraud. Newer methods of fraud, such as ”defrauding,” are exceptionally hard to detect, as traditional algorithms, such as rule-based algorithms or statistical methods, are ineffective. Traditional rule-based algorithms made by domain specialists hold for a period of time but are slow to change to newer fraud patterns. Furthermore, statistical methods such as mean, median, and standard deviation perform well in normal distributions but not well in fraud sets, as fraud sets, inherently, are excessively imbalanced causing them to struggle with high-dimensional data and non-normal datasets [[2], [4]].

Additionally, the Natural Security Memorandum 22 - 09, issued by the US government, highlights the growing threat of quantum computers to existing cryptography systems. As quantum technology advances, many systems in place will be ineffective, as it will be able to reach to a level that it can break sophisticated encryption methods. The report says that ”significant risks to the economic and national security of the United States,” the White House cautioned that a quantum computer of sufficient size and sophistication “will be capable of breaking much of the public-key cryptography used on digital systems across the United States and around the world” [5]

This report is mentioned not to highlight concern against using quantum computing as later in the report, the benefits of quantum computing and the course of action are mentioned. However, it is to show the current fraud techniques are becoming so potent that current methods and protection that was state of art before are otherwise ineffective now.

Furthermore, even many approaches using machine learning suffer due to the huge class imbalance in fraud detection is attributed to the vast majority of transactions being legitimate. This is also seen in my dataset where the legitimate transactions far surpassed by a multiple of 10 the amount of fraudulent transactions.

Thirdly, fraud has a huge monetary loss. According to a report by the Association of Certified Fraud Examiners (ACFE), traditional fraud methods are ineffective in dealing with fraud. The report shows that 84% of the fraudsters show a behavioral red flag and 43% of the fraud is only detected by insider tips [6]. These statistics further highlight the ineffectiveness and the classification of DEDE as traditional methods, are not efficiently able to handle the changing patterns of fraud, and are similarly not able to detect these red flags, and ultimately need help from insiders to detect fraud.

The report shows that an estimate of 5% of organizational revenue is lost due to fraud. It also shows that more than 5 trillion US dollars have been lost globally due to fraud, with 500 billion lost per year [6]. Because fraud has proved itself to surpass all conditions of DEDE, we consider it a DEDE.

II. LITERATURE REVIEW

Numerous projects using ML and QML have been attempted to create effective fraud detection systems. Trividea et al. (2020) aimed to create a more efficient credit card fraud detection system by proposing a machine learning feedback loop-based fraud detection. They tested several models such as Random Forests, Decision Trees, Artificial Neural Networks (ANN’s), Support Vector Machines (SVM), Naive Bayes, Logistic Regression, and Gradient Boosting, where they were able to show the random forests perform the best in terms of accuracy, recall, and accuracy. [7].

Illiberi et al. (2021) on the European credit card fraud dataset where Synthetic Minority Oversampling Technique (SMOTE) was use to respond to the high-class imbalance (the high rate of legitimate transactions to fraudulent transactions). Then Ada-boost was used in collaboration with other ML models, like decision trees, logistic regression, and XGBoost. They say that Random forests and XGBoost both had high accuracy of 90-99. [8].

Raghaven and Gayar (2019) explored the dynamic nature of fraud and how technological advancements allow fraudsters to bypass security systems. They benchmarked various machine learning methods, including K-nearest neighbors (KNN), random forest, and support vector machines (SVM), as well as deep learning models such as autoencoders, convolutional neural networks (CNN), restricted Boltzmann machines (RBM), and deep belief networks (DBN). They evaluated the models using the European, Australian, and German datasets, with Area Under the ROC Curve (AUC), Matthews Correlation Coefficient (MCC), and cost of failure as their metrics. [9]

Quantum Machine Learning (QML) has potential, but current hardware, like Noisy Intermediate-Scale Quantum (NISQ) systems, faces significant limitations due to noise and qubit constraints, as discussed by Preskill (2018). To overcome these challenges, hybrid quantum-classical models have been developed. For instance, Sakhnenko et al. (2021) introduced

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a Hybrid Classical-Quantum Autoencoder (HAE) that combines classical auto-encoders with quantum circuits, improving anomaly detection tasks through enhanced precision and recall. This approach demonstrates how hybrid systems can capitalize on quantum computing's future potential while compensating for present hardware limitations. [[10], [11]]

Furthermore, Grossi, Ibraham, Radescue et al. (2022), proposed a mixed quantum-classical method for fraud detection using quantum feature selection. This approach employs a quantum support vector machine (QSVM) algorithm for classification in the financial payment industry, integrating IBM Safer Payments with IBM Quantum Computers. By using a real-world data set of card payment transactions, they demonstrated that combining classical machine learning algorithms, such as random forest and XGBoost, with quantum algorithms significantly improves fraud detection accuracy. The study highlights the effectiveness of a hybrid approach recommending further exploration of such models to enhance unsupervised data models and fraud prevention techniques. [12]

III. Dataset and Methodology

Our research utilizes a suite of datasets created by Feedzai Research, designed specifically for fraud detection in bank accounts. The dataset suite includes six temporal variants, each synthetically generated to simulate real-world scenarios. The datasets span from February to September and contain 31 features with over 1 million instances per dataset. To ensure privacy, the dataset includes induced Laplacian noise, preserving the privacy of sensitive information while maintaining the utility for research purposes. [13].

A. Dataset Creation

The suite was originally created due to the lack of large-scale public datasets or suites to detect bank account fraud. It was created because other datasets or suites like the European credit card holder datasets were not very useful for researchers, as they have features like V1, V2, V3, V4...V20. These protected attributes aren't useful for researchers as they don't have a base value to ground their results. [13].

The dataset variants were intentionally designed to reflect common biases found in real-world fraud detection scenarios. For example, some datasets introduce group size disparity, where certain demographics, such as race, gender, or age groups, appear more likely to commit fraud. Other datasets feature prevalence disparity, simulating higher instances of fraud in specific regions or among certain demographics, such as older individuals. Additionally, temporal biases are introduced to mimic fraud patterns that spike during certain periods, such as holidays, where fraudulent activity tends to increase. These biases help analyze how fraud behavior shifts across different conditions. To simulate real-world challenges, missing values and induced Laplacian noise were also included to test how models handle incomplete and imperfect data. The authors purposefully created these biases to closely align with the complexities of actual fraud detection systems. The datasets were evaluated using the Pearson Correlation Coefficient, allowing for an analysis of feature relationships and helping to measure the effectiveness of fraud detection across different variants. [13].

B. Classical Methods

The dataset suite used in this project required minimal preprocessing due to XGBoost's built-in mechanism for handling missing values. Although the synthetic datasets contained missing data, this was an intentional design choice to simulate real-world conditions, where missing values often occur. Additionally, the dataset was augmented with Laplacian noise to further mimic practical scenarios. While synthetic data generally does not contain duplicates, we manually removed any duplicates and highly correlated features to enhance the dataset's integrity and ensure better model performance.

We conducted data analytics to observe the distribution of various features, utilizing histograms for numerical variables and bar graphs for categorical data. Categorical features were then encoded using level encoding to facilitate scatter plot creation, which enabled us to identify highly correlated features early in the process. This analysis was particularly useful for the quantum section of the project, as it allowed us to determine key features that contributed to performance.

The dataset was split into training and testing sets with an 80-20 ratio, providing ample data for model learning while reserving a portion for testing. The model was initialized using XGBoost, which effectively handled missing values. To optimize performance, we applied RandomSearchCV for hyper-parameter tuning, adjusting parameters such as step size, gamma, and max depth across 150 iterations with a 3-fold cross-validation. This tuning process enhanced the model's accuracy and ensured that it was neither under-fitting nor over-fitting.

The use of a synthetic dataset in this project aimed to replicate real-world conditions, where data often contains missing values due to errors or incomplete information. This approach allowed us to test quantum computing's potential in fraud detection under conditions similar to those faced in practical applications. By carefully managing the dataset's missing values and optimizing the model, we ensured that the dataset remained robust and the model reliable.

C. Quantum Methods

Due to current hardware limitations, significant preprocessing was necessary for the quantum portion of the code. The dataset contained six columns with missing data. Three columns with over 25% missing values were removed entirely, as Quantum Support Vector Machines (QSVMs) cannot efficiently handle missing values. The remaining three columns were addressed by removing rows with missing data. In addition to eliminating duplicates, we also removed unnecessary information. Categorical data was level-encoded into numerical data to ensure compatibility with the machine learning model.

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Initially, we intended to use 16 qubits to run the QSVM, but due to computational and time constraints, we reduced the feature set to eight qubits. This reduction, although necessary, limited the model's performance. For future projects, running the model for longer periods may allow for more accurate results. To train the model, we sampled 750 instances, 250 representing fraud cases and 500 non-fraud cases.

We applied a scalar to transform the feature training set and converted the target training set into a numerical array, followed by converting both sets to float types. With eight qubits, each representing a feature, we constructed a quantum kernel and circuit using Angle Encoding. We then applied the Support Vector Machine (SVM) classifier with the quantum kernel. This pre-processing and scaling process was repeated for the test set, which also consisted of 750 instances, and we ran the model six times to obtain reliable key performance indicators (KPIs) for both the training and testing sets.

IV. Results

After accumulating the result of both the quantum and classical setting, we show the scores of each variant based on precision, recall, F1, ROC.

TABLE I
XGBoost Classification Score Chart

Model	Precision	Recall	F1	ROC
Base	0.9900	0.9800	0.8800	0.8957
Variant I	0.9900	0.7900	0.8700	0.8919
Variant II	0.9900	0.8200	0.8900	0.9072
Variant III	0.9900	0.8800	0.9300	0.9572
Variant IV	0.9900	0.8100	0.8800	0.9000
Variant V	0.9900	0.8500	0.9100	0.9438

Fig. 2. XGBoost KPI Metric Scores

The results of the standard state-of-the-art models were very high on average, ranging from 89% to 95% ROC-AUC score. These models were tested on XGBoost. In Figure 1, the table displays different Key Performance Indicators (KPIs) for each variant of the models tested. These KPIs include metrics such as the F1 score, ROC-AUC score, precision, recall, and accuracy, providing a comprehensive evaluation of model performance.

Furthermore, the table highlights that measurements like the F1 score, the ROC-AUC, and the recall, which best represent the trade-off between false positives and false negatives, show high precision and a similarly high recall, the F1 and the ROC-AUC scoring. The F1 score, in particular, is a balanced mean of precision and recall, making it a balanced measure for evaluating models when there is an uneven class distribution.

In figure 3, the quantum model didn't do as well as the classical models and under-performed as compared to the XGBoost. Due to the limitations of the quantum kernel and limited number of samples, the quantum did a fair job despite all the restrictions. Future researchers should redo the experiment using better hardware and better methods. We can also tell where the quantum models suffer by training them all with the same quantum kernel. Variants II, III, IV, and V all have biases that decrease KPI metrics.

Quantum Classification Score Chart

Model	Precision	Recall	F1	ROC
Base	0.7732	0.8320	0.8015	0.6720
Variant I	0.7044	0.7100	0.7124	0.6580
Variant II	0.6243	0.5765	0.5612	0.5692
Variant III	0.6332	0.5513	0.5722	0.5474
Variant IV	0.6320	0.5900	0.6123	0.5870
Variant V	0.6123	0.5924	0.6120	0.5632

Fig. 3. XGBoost KPI Metric Scores

Multiple KPIs were used to track fraud detection due to its complexity. Using the F1 score, class imbalances could be handled by balancing precision and recall, ensuring that fraud detection is done while avoiding false positives. Recall indicates how many fraud cases were correctly identified. Precision indicates the proportion of correctly classified fraud cases. AUC-ROC distinguishes fraud from non-fraud classes as well. AUC, especially, as well as other methods, require a lot of training data. This means fewer hybrid scores are most likely due to limited training data, qubit usage, and class detection challenges.

Variant 1 shows a slight performance decrease, while Variants 2 and 3 experience a more significant drop. Variants 4 and 5 perform better than Variants 2 and 3 but still fall short compared to the base and Variant 1. Based on these scores, we can attribute certain factors to the biases introduced. Variant 1 has a group size disparity, meaning that the groups differ in size. Separability disparities and prevalence disparities are introduced in variants 2 through 5, which may contribute to performance declines. This could be due to challenges with generality and inference, likely caused by hardware limitations.

V. Discussion

As far as our research suggests, their is a possibility for future quantum integration to fraud detection systems.

The lower KPI scores for the quantum model are expected due to several factors. Firstly, the quantum model was trained on a limited number of samples, which is a significant constraint. Quantum machine learning models typically require substantial computational resources and large samples to achieve performance levels comparable to classical models. Additionally, the necessity to re-run the quantum kernel for each variant adds computational overhead. This re-running process is resource-intensive and can impact the overall performance of the model.

Quantum Computing is still in its early stages of development. The current hardware limitations of quantum computers can lead to lower performance. The scores reflect these limi-

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tations, highlighting the current challenges faced by quantum computing technology and its algorithms.

Despite the lower scores, the machine learning model still provides valuable insights into the potential and challenges of quantum machine learning. However, the ability of the quantum model to achieve any reasonable level of performance with limited data and computational resources is promising. This performance demonstrates the practicality of quantum models and their potential for future developments as the technology matures.

The high performance of the classical XGBoost model serves as a benchmark, setting a standard for comparison. The quantum model's lower scores highlight the areas needing improvement. The current scores provide a baseline for evaluating future progress in quantum machine learning. As quantum computing hardware and algorithms advance, it is expected that the gap between classical and quantum model performance will narrow. Future improvements in quantum computing technology could lead to significant enhancements in the performance and applicability of quantum machine learning models.

It is essential to acknowledge the limitations of this study. The biggest limitation is quantum hardware as computational resources are needed for widespread implementation. Furthermore, implementing the Hybrid Quantum-Classical solution on a corporation level might still need major technical improvements to fully implement. Addressing and improving technology is the only thing that can help us tackle these problems.

VI. Conclusion

Classical machine learning is a state-of-the-art technique to combat fraud, but it has limitations. As technology advances, so too must our methods. Many companies still suffer significant losses due to their reluctance to adopt artificial intelligence programs or their reliance on weak fraud detection systems. Although conventional machine learning and artificial intelligence are powerful, the accuracy and effectiveness of fraud detection can be significantly enhanced by incorporating quantum techniques and generated data to create abundant, privacy-protected synthetic data.

This study has demonstrated the significant potential of a hybrid quantum-classical approach for bank account fraud detection. Due to limited hardware, future experiments would be needed, to fully determine whether or not a hybrid setting is beneficial. This research is important because it paves the way for future innovations in fraud detection and potentially other fields. By bridging the gap between cutting-edge quantum computing and established machine learning techniques, this study advances the understanding and application of quantum-classical hybrid systems.

For future improvements, researchers should consider using synthetic data without missing values to better assess the full potential of quantum-enhanced methods. The use of Quantum GTANS (QCGTANS) models could further enhance the quality of synthetic data. Additionally, exploring more quantum algorithms in fraud detection, beyond the methods we used would be helpful. Future work should aim to develop quantum versions of classical machine learning algorithms like XGBoost to fully harness the power of quantum computing in fraud detection. One potential avenue is refining quantum encoding techniques to further improve performance and reduce computational demands. Another area of interest could be the application of this hybrid approach to other domains beyond financial fraud detection. Overall there are many avenues or directions quantum-based research can take but we hope our research can serve as a benchmark for future research.

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